

AIGR Hypothesis Review — HETGA *Cgas* (A0AAX6RS70)

Hypothesis (slug: `exogenous-dsrna-response-versus-dna-sensing`): *Heterocephalus glaber* Cgas participates in **cellular response to exogenous double-stranded RNA** (GO:0071360).
Focus type: function_assignment.

Executive Judgment

Verdict: Refuted / over-annotated (as a dsRNA response).

The seed hypothesis rests on a GO annotation that is a **term-selection error at the donor**, not a genuine dsRNA-response function. Mouse cGAS (Q8C6L5) carries GO:0071360 "cellular response to exogenous **dsRNA**" supported by four experimental references —

P 23258413

P 24077100

P 28214358

P 28363908

— and **every one of these papers assays double-stranded DNA (transfected DNA, DNA virus, λ/dsDNA binding), with no exogenous dsRNA / poly(I:C) experiment at all**. cGAS is the canonical cytosolic **dsDNA** sensor; it binds RNA structurally but is not catalytically activated by dsRNA, and even DNA:RNA hybrids are largely inert.

The HETGA annotation is **pure phylogenetic inheritance** (IEA:TreeGraft via PANTHER node PTN002579681; the source IBA is GO_REF:0000033). There is **no naked mole-rat experimental evidence** of any kind. Therefore this is **ambiguity/error in the donor evidence, not target-specific evolutionary divergence**.

Important caveat honoured from the seed: lack of direct RNA binding or poly(I:C)-stimulated catalysis does not, by itself, exclude an *indirect* cellular response to exogenous dsRNA (e.g., dsRNA-induced mitochondrial damage → mtDNA release → cGAS). **However, GO IDA/IMP**

evidence requires the cited experiment to demonstrate the process, and none of the four donor papers tests dsRNA. So even granting the indirect possibility, the cited evidence does not establish it.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
P 23258413 (Sun 2013, Science)	Direct assay + mutant/knockdown (IMP for GO:0071360)	Refutes dsRNA; supports dsDNA	Is cGAS a nucleic-acid sensor and of which?	"cGAS is a cytosolic DNA sensor that induces interferons by producing... cGAMP"; assays = DNA transfection & DNA virus	Mouse/human cells
P 24077100 (Ablasser 2013, Nature)	Direct assay (IDA for GO:0071360)	Refutes dsRNA; supports DNA/cGAMP	How does cGAS-derived immunity spread?	"Cytosolic DNA is sensed by... cGAS"; cGAMP transferred via gap junctions	Murine & human cells
P 28214358 (Lee 2017, FEBS Lett)	Biochemical/structural (IDA)	Refutes dsRNA; supports dsDNA	N-terminus effect on cGAS catalysis	"catalytic activity of core-cGAS decreased as the length of double-stranded DNA (dsDNA) increased"	In vitro / cells
P 28363908 (Tao 2017, J Immunol)	Single-molecule + biochem (IDA)	Refutes dsRNA; supports dsDNA	N-terminus role in DNA binding/activation	"binding of nonspecific dsDNA by the N-terminal domain of cGAS promotes its activation" (λ DNA)	Human cGAS, in vitro/cells
PANTHER PTN002579681 / GO_REF:0000033 (IBA)	Phylogenetic (ECO:0000318)	Qualifies/ competing	Ortholog-based propagation of donor terms	Propagates the (erroneous) dsRNA term across the cGAS	Computational

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
A0AAX6RS70 IEA:TreeGrafter (UniProt/ QuickGO)	Computational (IEA)	Qualifies	Is HETGA term experimentally grounded?	clade, incl. HETGA GO:0071360 on HETGA is IEA only; no experimental support	<i>H. glaber</i>
 42629426 (Lebrec 2026)	Review-adjacent primary tool paper	Supports dsDNA identity	cGAS ligand landscape	"cGAS is the primary innate immune DNA sensor"; "simple DNA:RNA hybrids are inert "	Human cells

GO Curation Implications

Lead (requires curator verification):

- **Do not assign GO:0071360 (cellular response to exogenous dsRNA) to HETGA *Cgas*.** The IEA:TreeGrafter propagation inherits an unsupported donor term.
- **Flag the donor annotation (mouse Q8C6L5 → GO:0071360) for correction.** All four cited experiments demonstrate response to exogenous/cytosolic **dsDNA**, not dsRNA. The biologically correct BP analog already exists: **GO:1990786 "cellular response to dsDNA"** (aspect BP). This is best treated as a **dsRNA↔dsDNA term-selection error**.
- Retained/positive terms that *are* well supported for cGAS and appropriately propagated to HETGA: **GO:0061501 (2',3'-cGAMP synthase activity, MF)**, **GO:0003690 (dsDNA binding, MF)**, **GO:0140896 (cGAS/STING signaling, BP)**, **GO:0032481 (positive regulation of type I IFN production, BP)**, **GO:0002230 (positive regulation of defense response to virus by host, BP)**. These, not a dsRNA term, capture the gene product's function.

- Avoid "protein binding" as a recommendation — more informative MF/BP terms above are supported.

Mechanistic Scope

Immediate molecular function tested by the donor references: binding of cytosolic **double-stranded DNA** by cGAS and DNA-stimulated catalysis of **2',3'-cGAMP** (a nucleotidyltransferase reaction, EC 2.7.7.86), which activates STING → TBK1/IRF3 → type I IFN. The N-terminal papers (28214358, 28363908) dissect how the low-complexity N-terminus modulates **dsDNA** binding and catalytic output.

- **Direct gene-product activity:** dsDNA sensing + cGAMP synthesis.
- **Downstream (not the tested MF):** type I IFN induction, antiviral state, paracrine cGAMP spread, senescence.
- **Not demonstrated by any cited paper:** a response to exogenous **dsRNA** — neither direct (RNA binding/poly(I:C) catalysis) nor indirect (dsRNA-triggered mtDNA release feeding cGAS).

Conflicts and Alternatives

- **Term mislabel (most likely):** "dsRNA" was selected where "dsDNA" was meant; the four references and the entire cGAS field describe DNA sensing. Analog term GO:1990786 exists.
- **Database carry-over:** the error is amplified by IBA (PANTHER PTN002579681) → IEA:TreeGrafter to orthologs including HETGA, so multiple species now carry it.
- **Indirect-response steelman:** exogenous dsRNA can cause mitochondrial/nuclear damage that releases DNA activating cGAS; this *could* justify a broad "response to dsRNA" chain, but it is (a) not what the cited papers test and (b) would still be indirect and better annotated to the DNA-sensing MF/BP with the actual trigger, not a direct dsRNA term.
- **No paralog confusion:** the donor is bona fide cGAS/MB21D1; the issue is term choice, not gene identity. HETGA A0AAX6RS70 is a clear cGAS ortholog (XP_021099394.1, Mb21d1).

Knowledge Gaps

1. **Was the donor "dsRNA" term deliberate or a typo for "dsDNA"?** Checked: all 4 cited abstracts test DNA; a DNA analog term exists. Matters because it determines whether to correct vs remove. Resolve by inspecting the original UniProt/MGI curation record and the exact figure cited.
 2. **Any *H. glaber*-specific cGAS immune data?** Checked: no experimental UniProt evidence; HETGA term is IEA-only. Matters for whether species divergence could rescue the term. Resolve with naked mole-rat cGAS ligand/activation assays (poly(I:C) vs dsDNA).
 3. **Does naked mole-rat cGAS have altered nucleic-acid selectivity?** Not established here. Resolve with recombinant HETGA cGAS in vitro cGAMP assays across dsDNA, dsRNA, and RNA:DNA hybrids.
-

Discriminating Tests

- **In vitro cGAMP synthesis assay** with purified HETGA (and mouse) cGAS challenged with dsDNA vs poly(I:C)/dsRNA vs RNA:DNA hybrids — directly separates "dsDNA sensor" from "dsRNA responder." Expectation: activation by dsDNA only.
 - **Cellular IFN- β reporter** in HETGA cells: transfected dsDNA vs poly(I:C), $\pm$ cGAS knockdown/knockout, to test whether any dsRNA-driven IFN is cGAS-dependent (would test the indirect route).
 - **Curation provenance check:** retrieve the exact experiment/figure the donor annotation cites for GO:0071360 to confirm dsRNA vs dsDNA wording.
-

Curation Leads (require curator verification)

- **Action:** Remove/replace GO:0071360 on HETGA *Cgas* (A0AAX6RS70); it is IEA:TreeGrafter inheritance of an unsupported donor term.
- **Candidate replacement term:** GO:1990786 "cellular response to dsDNA" (BP) if a stimulus-response BP is desired; otherwise rely on the well-supported GO:0140896 cGAS/STING signaling and GO:0032481 positive regulation of type I IFN production.

- **Upstream fix:** File a note to correct mouse Q8C6L5 GO:0071360 (dsRNA→dsDNA) and re-propagate; PANTHER node **PTN002579681** should be updated so all clade orthologs stop inheriting the dsRNA term.
- **Candidate references + verify-snippets:**
 - **P 23258413**
— "cGAS is a cytosolic DNA sensor that induces interferons by producing the second messenger cGAMP."
 - **P 28214358**
— "the catalytic activity of core-cGAS decreased as the length of double-stranded DNA (dsDNA) increased."
 - **P 28363908**
— "binding of nonspecific dsDNA by the N-terminal domain of cGAS promotes its activation."
 - **P 24077100**
— "Cytosolic DNA is sensed by the receptor cyclic GMP-AMP (cGAMP) synthase (cGAS)."
- **Suggested question for curators:** Is the donor GO:0071360 traceable to a specific dsRNA experiment, or is it a dsDNA/dsRNA wording error? If the latter, correct at source before any ortholog carries it.

Provenance / Artifacts

Computed provenance from QuickGO/UniProt REST queries (run 2026-09-21):

- `artifacts/go0071360_annotation_provenance.csv` — every GO:0071360 annotation on mouse Q8C6L5 and HETGA A0AAX6RS70 with qualifier/evidence/reference. **All are positive involved_in (not NOT) annotations**; mouse = 1 IBA + 1 IMP + 3 IDA; HETGA = single IEA:TreeGrafter (GO_REF:0000118).
- `artifacts/evidence_matrix.csv` — the evidence matrix above in tabular form.
- `artifacts/go_decision_table.csv` — GO decision table (remove/replace/retain per term).

Key confirmation this iteration: the mouse and HETGA annotations use qualifier `involved_in` — i.e., the database positively asserts participation in exogenous-dsRNA response, so this is a genuine over-annotation to correct, not a mis-read `NOT` (does-not-respond) annotation.

Limitations

- Analysis is annotation/literature-based; abstract-level reading of the four references (full texts not machine-parsed here) — but the abstracts are explicit about DNA. No new wet-lab data. QuickGO/UniProt snapshots as of the run date (2026-09-21).

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery